

Edith Gu

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EDUCATION

University of California San Diego

Master of Science, Data Science, GPA: 3.9 / 4.0

San Diego, CA

Sep. 2025 - Mar. 2027

Zhongnan University of Economics and Law

Bachelor of Science, Data Science, GPA: 3.9 / 4.0

Wuhan, China

Sep 2021 - Jun 2025

PROFESSIONAL EXPERIENCE

AskTuring.ai

AI Engineer Intern

San Diego, CA, USA

Mar. 2026 - May 2026

- Built private-knowledge Q&A for **50** financial-industry clients (banks, asset managers, PE firms) that lets enterprises upload documents, developing the **RAG** pipeline and agentic **workflows** on **AWS** with **Bedrock** hosted Claude for generation so research, compliance, and client-service teams can query their own data.
- Designed partial-failure handling for a **6-role agentic** workflow (**LangGraph**, Pydantic-typed **tool calling**), adding mandatory completion state/missing-item list per agent that Planner consumes for retry and SQL-to-document fallback.
- Developed an **LLM-as-a-judge** evaluation framework (correctness, citation accuracy, refusal rate, recall@k) on a **300-question** analyst-labeled golden set, run in **LangSmith** as the pre-release regression gate and paired with **Prometheus/Grafana** production dashboards, driving accuracy from **68 to 89** out of 100.

Hubei Ronghai Investment Co., Ltd.

Software Engineer Intern

Wuhan, China

Jun. 2025 - Aug. 2025

- Reduced p99 latency of financing-application submission from **2.1s to 300ms** during promotional windows on a supply chain finance platform serving **600** suppliers, offloading risk-control and notification calls to **RocketMQ** consumers with idempotent handlers and pre-warming **Redis** with campaign quotas.
- Eliminated duplicate disbursements caused by gateway-level retries after upstream timeouts in the disbursement service's **Java Spring Boot REST API**, extending idempotency keys to the full business flow ID and persisting processed-request records to **MySQL**, achieving **zero** duplicates across **4,000** disbursements.
- Containerized the financing-application and disbursement services with **Docker**, deployed to **Kubernetes** with rolling updates and HPA autoscaling for promotional traffic spikes, and set up **GitHub Actions CI/CD** running unit and **JMeter** load tests as a release gate.

Wuhan FraserGen Co., Ltd.

Software Engineer Intern

Wuhan, China

Feb. 2024 - Mar. 2024

- Engineered an internal AI assistant for genomics research workflows using **LangChain**, **FastAPI**, and **WebSockets**, integrating **GPT-4-Turbo** for domain reasoning and **Coqui TTS** for real-time speech output to support natural-language and voice interaction across research teams.
- Built a **RAG pipeline** over internal genomics documents using **OpenAI text-embedding-3-large** and **Qdrant**, reducing hallucinated answers by **35 points** in internal evaluation and improving factual accuracy in scientific responses
- Applied recursive and semantic chunking to long, structured biological documents, tuning retrieval granularity to reduce embedding redundancy and improve retrieval precision for complex genomics queries.

PROJECTS

MedScribe — On-premise medical speech-to-note assistant

2026

- Built a fully on-premise clinical dictation tool in **Python**, streaming **Whisper-large-v3** inference in **PyTorch** with **FastAPI + WebSocket** streaming, feeding a locally served **Qwen2.5-7B (vLLM)**, with terminology correction via a local medical dictionary indexed in **Qdrant**, so no audio or notes leave the machine.
- Fine-tuned Whisper with **LoRA (PyTorch, HuggingFace Transformers + PEFT)** on **40 h** of medical dialogue and **25 h** of non-native-accented speech, reducing error rate on medical terms from **18% to 11%** and on accented speech from **15% to 9%** versus base Whisper, evaluated with **jiwer** on a held-out, per-accent / per-terminology test split.

TECHNICAL SKILLS

- **AI/ML**: Machine Learning (Logistic Regression, SHAP), LLM Applications (GPT-5.5, RAG), OpenAI Embeddings, Vector Databases (Pinecone, OpenSearch), LangChain, LlamaIndex, MCP (Model Context Protocol), LLM-as-a-judge, Few-shot Prompting, Attribution Modeling
- **Languages & Frameworks**: Python, FastAPI, Ruby on Rails, TypeScript, Java (Spring Boot), C# / .NET Core, React, JavaScript, Node.js
- **Databases**: PostgreSQL, MySQL, Redis, Azure SQL, BigQuery, MongoDB, Vector Databases (Pinecone, OpenSearch)
- **Cloud**: AWS (Multi-Region, API Gateway, SQS, Lambda), Azure (Log Analytics/Kusto, Cloud Services), CI/CD, Kubernetes, Docker
- **APIs & Protocols**: REST APIs, WebSocket, OAuth 2.0
- **Monitoring**: Prometheus, Grafana, Azure Log Analytics, Distributed Tracing, LLM Observability